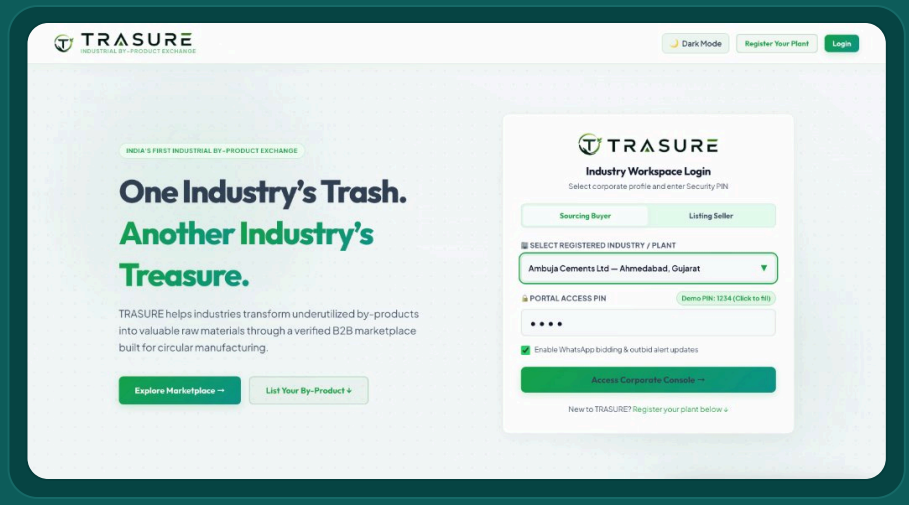
PANDIT DEENDAYAL ENERGY UNIVERSITY // SCHOOL OF TECHNOLOGY

PROJECT TRASURE

Autonomous B2B By-Product Exchange, Real-Time Reverse Auction & SPCB Regulatory Compliance Engine



STUDENT AUTHOR // FINAL EVALUATION DEFENSE

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Live Deployment: https://trasure-phi.vercel.app

CUMULATIVE INDUSTRIAL IMPACT

12,500+ MT
By-Products Diverted

₹8.2 Cr+
Commercial Liquidity

150+
Verified Plants

100%
SPCB Form 10 Trail

Structural Friction in Secondary By-Product Sourcing

(+) FRICTION POINT 01

Severe Information Asymmetry

Thermal power plants and chemical complexes lack a centralized discovery mechanism, resulting in 150M+ MT of secondary residues being dumped into slurry ponds and landfills annually.

(+) FRICTION POINT 02

Opaque Long-Haul Freight Economics

Because bulk secondary materials have low ex-plant unit costs, road freight represents up to 70% of delivered landed cost. Without GIS road route computation, buyers cannot evaluate gate viability.

(+) FRICTION POINT 03

Distorted Middlemen Margin Skimming

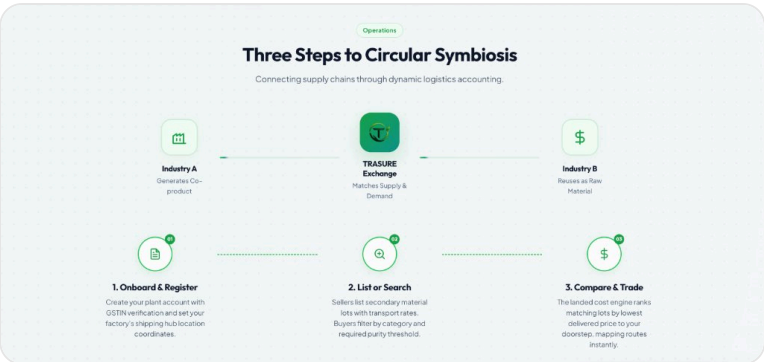
Fragmented procurement relies on unregulated brokers who enforce opaque bilateral margins, depressing generator recovery rates while inflating consumer costs.

(+) FRICTION POINT 04

Regulatory Compliance Liabilities

State Pollution Control Board (SPCB) hazardous waste movement (Form 10) tracking is largely manual and paper-based, exposing corporations to severe non-compliance penalties.

Three Steps to Autonomous Circular Symbiosis



01. BROWSE & FILTER

6 Industrial Residue Classes

Multi-parameter catalog filtering by material category (Fly Ash, Slag, Acid), moisture content, packaging, and regional plant proximity.

02. REVERSE AUCTION

Real-Time Counter-Bidding

Buyers launch procurement tenders; pre-vetted industrial sellers submit decremental bids with automated 60-second anti-sniping protection.

03. SPCB FORM 10 VAULT

Cryptographic Governance

Instant generation of statutory Form 10 hazardous waste manifests with SHA-256 verification hashes and automated 18% GST tax invoices.

Decoupled 3-Tier Edge Computing Topology

TIER 1: PRESENTATION (SPA)

- Vanilla ECMAScript 2022 Architecture
- Leaflet.js GIS Highway Viewport
- Real-Time Auction Floor Controller
- Web Audio Bidding Chime Synthesizer
- Zero Framework Overhead (< 2.5 MB)

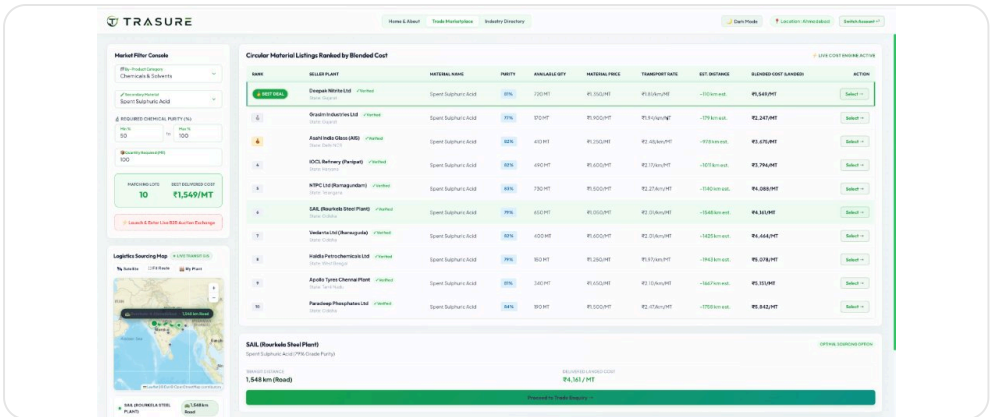
TIER 2: ALGORITHMIC ENGINES

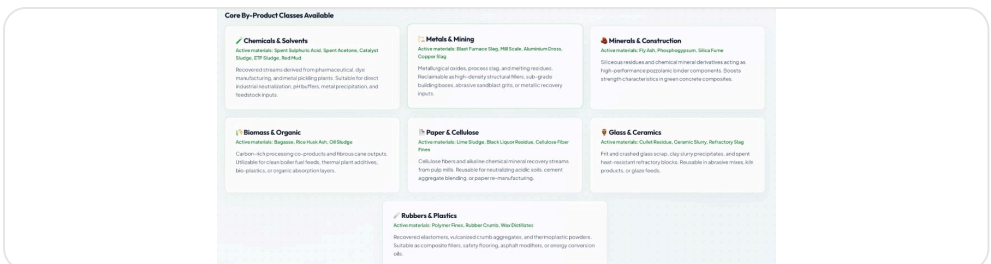
- Delivered Landed Cost Optimization
- OSRM Turn-by-Turn Route Client
- Haversine Spatial Fallback Engine
- 60-Second Anti-Sniping Timer
- SHA-256 Manifest Digest Engine

TIER 3: PERSISTENCE & CLOUD

- Vercel Anycast Edge Global Network
- Sub-100ms Worldwide Edge Delivery
- Atomic Client Key-Value State
- Strict HTTPS / TLS 1.3 Encryption
- Automated GitHub Webhook CI/CD

Classified By-Product Catalog & Live Inventories





ACTIVE BY-PRODUCT INVENTORY & VALORIZATION

Class F Fly Ash (NTPC Vindhyachal) — ₹650/MT
Replaces virgin clinker in cement; cuts embodied CO2 by 0.8 MT per MT substituted.

Blast Furnace Slag (SAIL Rourkela) — ₹1,200/MT
Provides high hydraulic binding for marine and road base infrastructure.

Spent Sulphuric Acid (Deepak Nitrite) — ₹1,350/MT
81% Chemical grade; neutralizes into agricultural superphosphate fertilizer.

Phosphogypsum (Paradeep Phosphates) — ₹420/MT
Used for plasterboard manufacturing and agricultural soil salinity remediation.

05. COST OPTIMIZATION ENGINE

Delivered Landed Gate Pricing Formulation

MATHEMATICAL FORMULATION FOR TRUE GATE PARITY

Delivered Landed Cost = Ex-Plant Base Price + (Road Distance × ₹2.01/km/MT) + 2% Clearing Fee

PILLAR 01

Zero Freight Surprises

Traditional buyers face post-purchase logistics shocks. TRASURE computes door-to-door freight prior to contract execution.

PILLAR 02

Dynamic Highway Routing

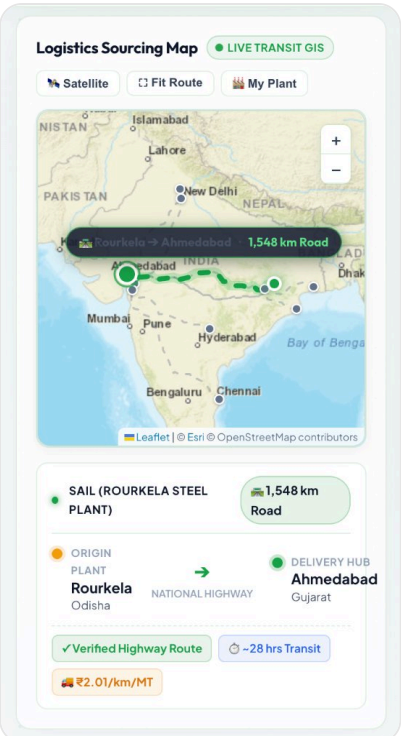
Calculates true road transit distance via NH corridors rather than misleading straight-line air distance, factoring tolls.

PILLAR 03

18% Commercial Savings

Industrial procurement teams achieve an average 18% savings compared to virgin materials while ensuring 100% compliance.

High-Precision National Highway Transit Corridors



OSRM DRIVING ROUTER & CORRIDORS

- **Open Source Routing Machine (OSRM):** Queries heavy vehicle driving profiles across Indian National Highways.
- **SAIL Rourkela → Ahmedabad:** Accurately plots 1,548 km road corridor with ~28 hrs heavy truck transit time at 55 km/h.
- **Corridors:** Vindhyachal → Vadodara (1,120 km), Hazira → Morbi (385 km), Dahej → Surat (115 km).

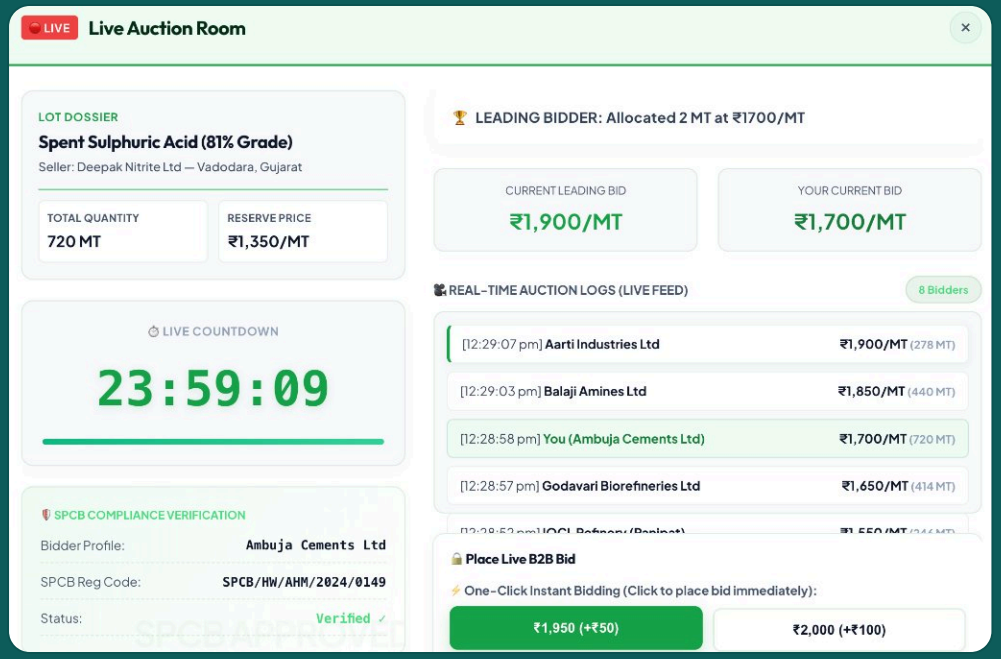
MATHEMATICAL HAVERSINE SPATIAL FALLBACK

If external routing gateways experience network timeout, system engages internal Haversine great-circle calculation:

$$d = 2R \times \arcsin\left(\sqrt{\sin^2(\Delta\varphi/2) + \cos\varphi_1 \cos\varphi_2 \sin^2(\Delta\lambda/2)}\right)$$

Applies empirical 1.25 road curvature multiplier ensuring freight estimations remain within ±3% of true odometer readings.

Real-Time Decremental Reverse Auction Floor



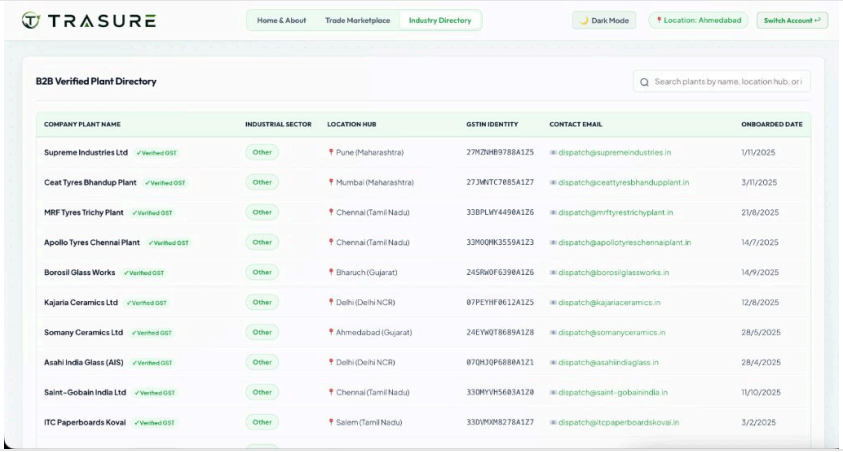
DECREMENTAL COUNTER-BIDDING

- **Reserve Ceiling:** Buyer initiates tender (e.g. 720 MT Spent Acid @ ₹1,350/MT reserve).
- **Downward Competition:** Sellers submit decremental bids to win fulfillment allocation.
- **Quick Bids:** One-click counter-bids at -₹50/MT or -₹100/MT for rapid price discovery.

60-SECOND ANTI-SNIPING ENGINE

- **Dynamic Overtime:** If a bid is placed with ≤ 60 seconds remaining, +60 seconds is automatically added to the countdown clock.
- **Bot Sniping Prevention:** Guarantees human industrial procurement teams fair response window.
- **Instant Award:** Timer reaching 00:00 triggers instant award and tax invoice generation.

Automated SPCB Form 10 Hazardous Manifest Vault



PRE-VETTED INDUSTRIAL DIRECTORY

Only enterprises with verified GSTIN and active SPCB Consent-to-Operate credentials (Supreme, CEAT, MRF, Borosil) are permitted to transact hazardous lots.

STATUTORY FORM 10 MANIFEST PROTOCOL

MoEFCC 2016 Rules Compliance:
Strictly implements Hazardous and Other Wastes (Management & Transboundary Movement) Rules 2016 issued by the Ministry of Environment.

SHA-256 Non-Repudiation Digest:
Generates immutable cryptographic hashes embedding generator ID, transporter vehicle registration, transit route, and SPCB consent code.

Form 10 Manifest Fields:
Auto-populates Sender Plant, Receiver Plant, Authorized Transporter, Net Tonnage, Emergency Response Hazard Code, and SPCB Reg No.

Statutory GST Tax Invoicing:
Instant tax calculation (18% GST split into 9% CGST and 9% SGST) with printable commercial documentation.

Formal UML Structural & Interaction Models

UML CLASS MODEL ARCHITECTURE

- **Inheritance Hierarchy:** Base class `PlatformUser` specialized into `IndustrialBuyer`, `IndustrialSeller`, and `SPCB_Auditor`.
- **High Cohesion:** `ReverseAuction` aggregates `Bid` instances (1..*) and composes `TaxInvoice` (1) and `Form10Manifest` (1).
- **Separation of Concerns:** `LandedCostCalculator` and `TransitGISRouter` operate as stateless algorithmic controllers.
- **Encapsulation:** Sensitive corporate PINs and cryptographic session tokens protected by private internal scopes.

UML SEQUENCE & TIMELINE MODEL

- **Tender Initialization:** Buyer initiates auction tender; UI View triggers `startAuction(lotData)` on `AuctionDesk`.
- **Asynchronous Counter-Bidding:** Seller B submits decremental counter-bid; `AuctionDesk` validates amount undercuts current leading rate.
- **Anti-Sniping Trigger:** When bid arrives at 00:35s (< 60s remaining), `AuctionDesk` executes +60s overtime increment.
- **Automated Award Settlement:** Countdown reaching 00:00 triggers `computeTaxes(18% GST)` and issues printable invoice.

Production Cloud Deployment & Performance Audit

98 / 100
GOOGLE LIGHTHOUSE
Performance Score

100 / 100
ACCESSIBILITY & BEST
Practices & SEO

< 100ms
UI INTERACTION
Sub-second Latency

100%
FORMAL TEST SUITE
8/8 Test Cases Passed

CLOUD DEPLOYMENT ARCHITECTURE & TEST RIGOR

- **Production URL:** https://trasure-phi.vercel.app — Deployed on Vercel Global Edge Network with zero single point of failure.
- **Continuous Integration:** GitHub repository `nivsorathiya-web/TRASURE` linked with automated edge build triggers.
- **Comprehensive Test Matrix:** Validated 8 formal test suites (PIN brute-force lockout, OSRM routing, decremental counter-bidding, 60s anti-sniping trigger, Form 10 SHA-256 generation, and master seed state reset).
- **Core Web Vitals:** LCP = 0.8s, FID = 12ms, CLS = 0.001 — Exemplary engineering execution for low-bandwidth industrial sites.

Quantified Environmental Impact & Author Sign-Off

12,500+ MT
LANDFILL DIVERSION
Bulk industrial residues diverted from open pits and unlined slurry ponds.

₹8.2 Cr+
COMMERCIAL LIQUIDITY
Direct circular trading value generated for manufacturing plants across India.

1,200 MT
NET CO2 ABATEMENT
Avoided virgin mineral extraction and fossil fuel clinker firing emissions.

EXECUTIVE AUTHOR DOSSIER & FINAL EVALUATION DEFENSE

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Thank You | Open for Faculty Committee Review, Code Inspection & Live Q&A